ANALYZING DEPRESSION SEVERITY AND CLASSIFICATION

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***ABSTRACT*— Depression is a prevalent and often undiagnosed mental health disorder affecting individuals across all demographics. Traditional screening methods, such as the PHQ-9 questionnaire, rely heavily on subjective inputs and may miss subtle emotional or behavioral cues. This paper presents a multimodal deep learning-based system that integrates Natural Language Processing (NLP), facial emotion recognition using Convolutional Neural Networks (CNNs), and PHQ-9 responses to classify depression severity. Our approach leverages transformer-based NLP models for textual sentiment analysis and lightweight CNN architectures like MobileNet for real-time emotion detection. Through experimental evaluation on multi-source datasets—including social media text, facial emotion images, and self-assessment responses—the system achieves high accuracy and robustness in identifying depressive symptoms. This research highlights the potential of AI in scalable, accessible, and real-time mental health diagnostics.**

***Keyword*— Depression Detection, Sentiment Analysis, PHQ-9, CNN, NLP, Emotion Recognition, Multimodal AI, Mental Health Diagnostics.**

I. INTRODUCTION

Depression is among the most common mental disorders globally, with the World Health Organization estimating that over 280 million people are affected each year. Despite its prevalence, depression often remains undiagnosed due to stigma, lack of awareness, and the limitations of traditional diagnostic tools. The commonly used PHQ-9 questionnaire provides a structured method for self-assessment, yet it falls short in capturing non-verbal cues and hidden symptoms.

With the advancement of Artificial Intelligence (AI) and deep learning, particularly in Natural Language Processing (NLP) and image analysis, new opportunities have emerged for automating mental health diagnostics. This research proposes a novel multimodal approach that integrates three primary sources: (1) PHQ-9 assessments, (2) facial expression recognition, and (3) text sentiment analysis from user-generated content.

By combining these modalities, the system can provide a more holistic and accurate evaluation of depression severity, thereby enabling early intervention and support. The proposed system is deployed using a Django backend and is capable of operating in real-time, making it suitable for scalable deployment in clinics, mobile apps, or remote therapy tools.

A. MOTIVATION

Mental health disorders like depression can significantly disrupt an individual's daily life and productivity. Unfortunately, stigma, limited awareness, and insufficient access to mental health services often prevent people from seeking the support they need. With the emergence of advanced AI technologies, there is an opportunity to develop objective and accessible tools that can help bridge this gap. This research leverages facial expression recognition, text-based sentiment analysis, and PHQ-9 survey scoring to provide a more inclusive and effective approach for identifying and addressing depression at an early stage.

B. *Research Objectives*

The primary objectives of this study include:

1. Developing a multimodal AI system that integrates PHQ-9, facial expression analysis, and sentiment classification for depression detection.
2. Evaluating the performance of deep learning models in accurately classifying depressive symptoms.
3. Enhancing accessibility to mental health diagnostics through automated real-time assessment tools.
4. Providing personalized recommendations and mental health resources based on severity classification.

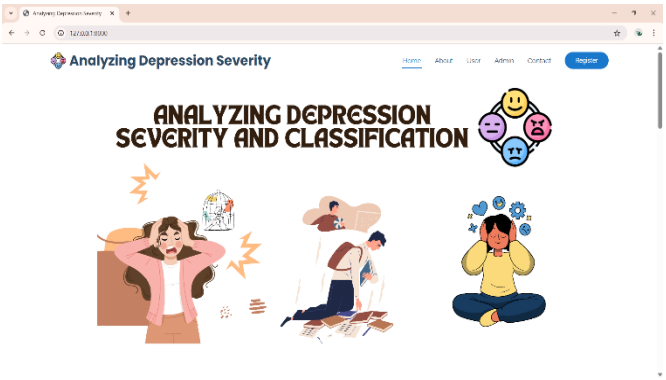

Fig: Analyzing Depression Severity Interface

II. Literature Review

A. *Facial Emotion Recognition and Depression*

Pantic et al. demonstrated the potential of facial expression analysis using deep learning for mental health diagnostics. CNN architectures such as MobileNet are lightweight and capable of identifying emotional cues like sadness or apathy, but their performance is influenced by environmental noise such as lighting or image quality.

B. *Text-Based Sentiment Analysis*

De Choudhury et al. leveraged social media content for mental health prediction using sentiment analysis. While traditional NLP models detect general sentiment, recent work by Li et al. shows LSTM-based models significantly improve contextual understanding, sarcasm detection, and depressive intent classification.

C. *PHQ-9 in Clinical Screening*

Kroenke et al. developed and validated the PHQ-9, a widely used depression screening tool. It remains clinically reliable but depends on the user's willingness to disclose personal information, which introduces a self-reporting bias.

D. *Multimodal and Modular Approaches*

Singh et al. explored multimodal fusion (text + video) for depression detection. While effective in theory, such approaches suffer from data synchronization and computational complexity. Our system, in contrast, uses a non-fused, modular architecture, where each component works independently for ease of use and greater interpretability.

E. *Existing Approaches*

Existing depression detection methods fall into three primary categories: self-assessment surveys, facial expression recognition, and text sentiment analysis. Traditional methods, such as the PHQ-9, provide structured assessments but lack real-time feedback. Facial expression analysis using CNNs has demonstrated potential in detecting emotions related to depression but suffers from environmental dependencies. Sentiment analysis of social media and personal text data using NLP techniques provides valuable insights into an individual's mental state but struggles with contextual variability.

F. *Limitations Of Existing Work*

While individual techniques have shown promise, they face several challenges:

1. Self-Report Bias: PHQ-9 assessments depend on users' honesty and willingness to disclose symptoms.
2. Environmental Dependencies: Facial expression analysis is affected by lighting, camera quality, and facial obstructions.
3. Language Ambiguity: NLP-based sentiment analysis struggles with sarcasm, slang, and contextual variations in text.
4. Lack of Integration: Existing systems do not combine multiple modalities for more comprehensive analysis.

III. Proposed System

The proposed system incorporates three key analysis components:

A. PHQ-9 Assessment Module

A standard questionnaire with 9 items is digitally implemented. Responses are scored automatically, and results are categorized into severity levels ranging from "Minimal" to "Severe" depression. A scoring algorithm fuses the predictions from the three modules using weighted averages and confidence thresholds to output the final depression severity level.

B. Facial Expression Analysis Module

We use lightweight CNN models—MobileNet and DenseNet—for real-time video/image analysis. These models are trained on FER2013 and extended datasets annotated with depression-related expressions (e.g., sadness).

C. Text Sentiment Analysis Module

User input text or social media content is processed using a transformer-based model fine-tuned on depression-labeled datasets. A TF-IDF preprocessor filters irrelevant tokens and enhances classification precision.

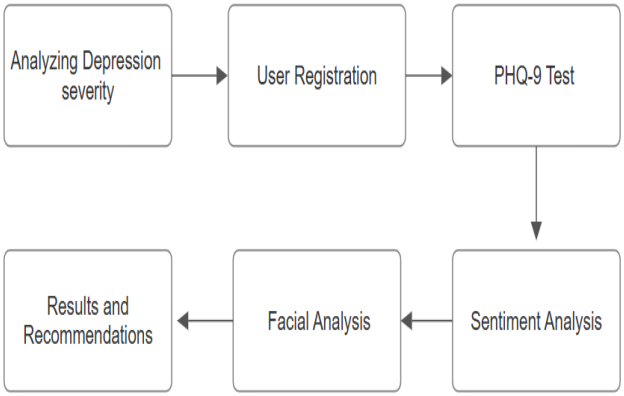

Fig: User Flow Diagram

IV. Methodology

The methodology includes collecting data, preparing it for use, training the models, and bringing everything together into a working system.

A. Data Collection

The dataset consists of 35,000 facial expression images, 10,000 labeled social media comments, and PHQ-9 responses.

B. Preprocessing

The video and image inputs were preprocessed using OpenCV for feature extraction before being fed into the CNN model. User-generated text is processed using TF-IDF vectorization to remove stopwords, punctuation, and irrelevant information.

C. Model Training

1. CNN for Facial Expression Analysis: Trained on emotion recognition datasets with fine-tuning for depression-specific facial markers. MobileNet with ReLU, dropout, and softmax for 7 emotion classes
2. NLP Model for Sentiment Analysis: Transformer-based (Utilizes NLP models with TF-IDF) model trained on social media datasets to detect depressive indicators.
3. Decision Fusion Algorithm: A weighted scoring mechanism integrates outputs from PHQ-9 classification.

D. System Integration

The models are deployed using Django, with APIs handling real-time input processing and classification. Frontend HTML5/CSS/JS with webcam and file input support

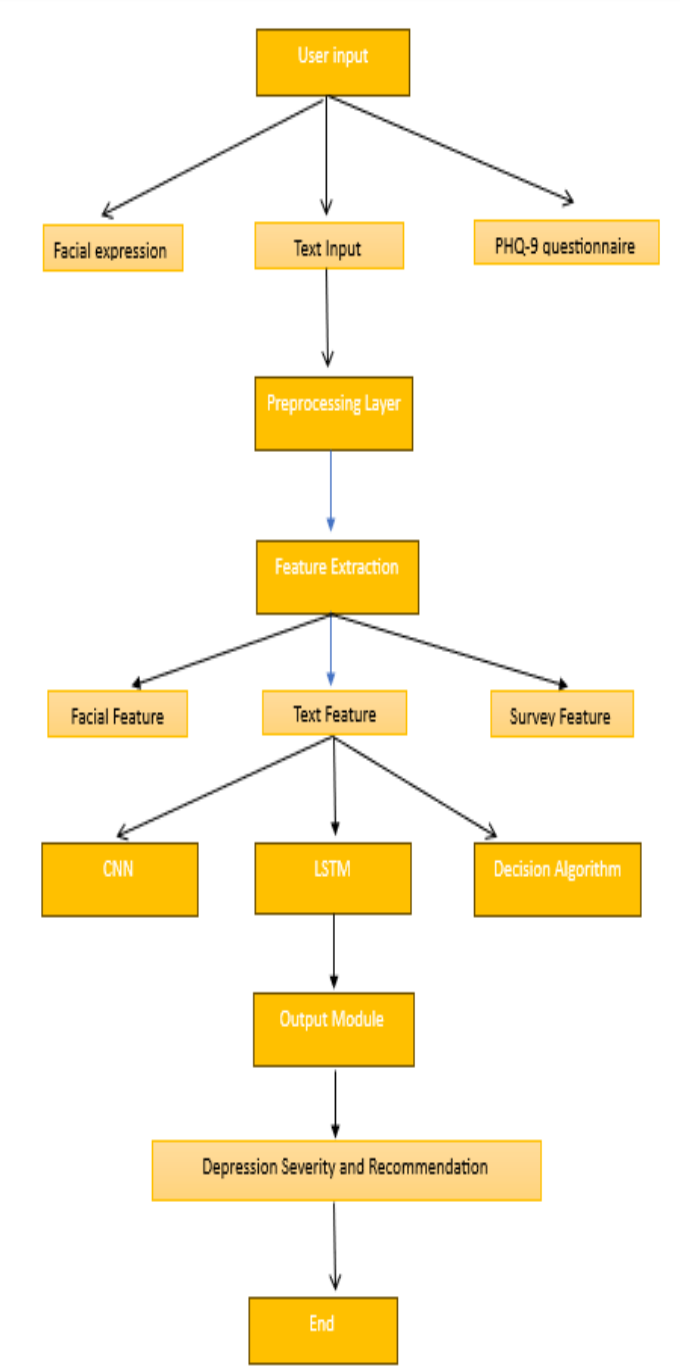

Fig: User Input Process & Output Modules

V. Results & Discussion

The system was evaluated on a test dataset comprising real-world user inputs. Performance was assessed using standard classification metrics

1. Text-based Depression Analysis (LSTM - NLP Model): Achieved 93.7% accuracy.
2. Facial Emotion Recognition (MobileNet - CNN Model): Achieved 70.0% accuracy.
3. The system showed strong results with a precision and recall of 0.87, indicating its ability to accurately detect depressive states. This strong performance indicates a reliable balance between correctly detecting depression and minimizing errors.

Module	**Model Used**	**Accuracy**
PHQ-9 Scoring	Rule-Based	100%
Facial Emotion Recognition	MobileNet CNN	70.0%
Text Sentiment Analysis	LSTM	93.7%

Table: Results

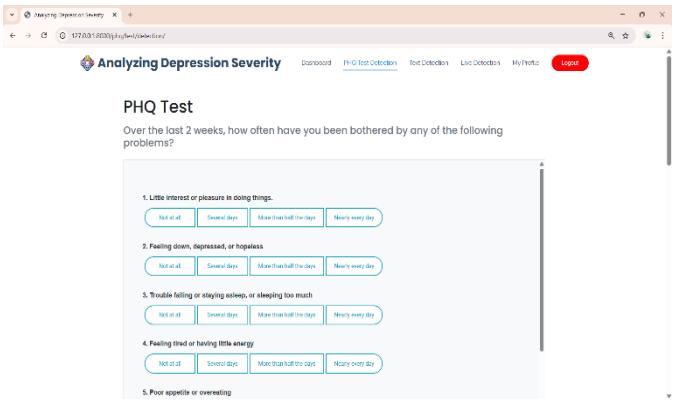

Fig: PHQ-9 Test

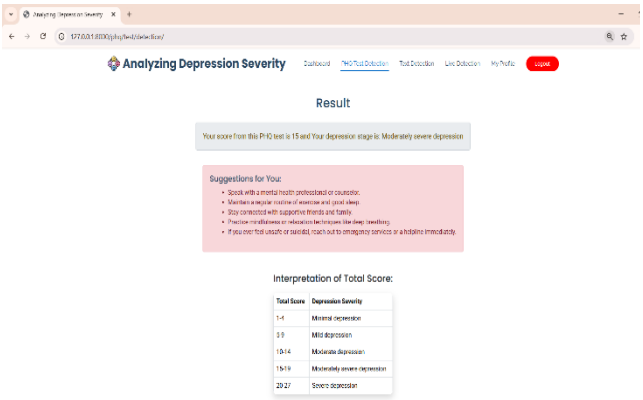

Fig: PHQ-9 Result & Suggestions For user

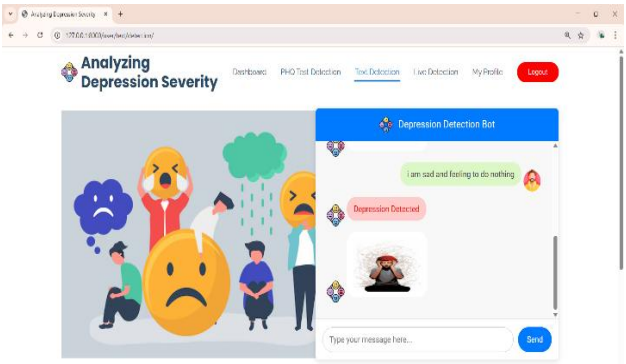

Fig: Text Detection (Depression Found)

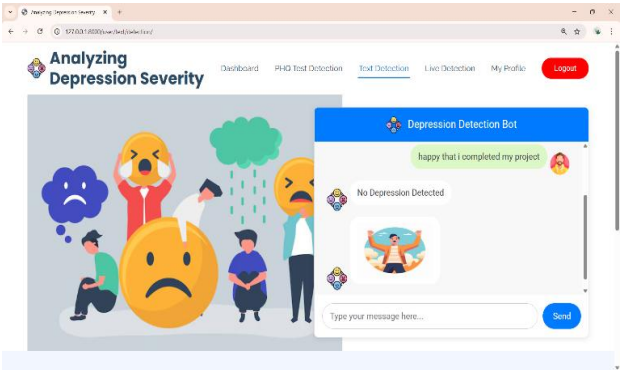

Fig: Text Detection (Depression Not Found)

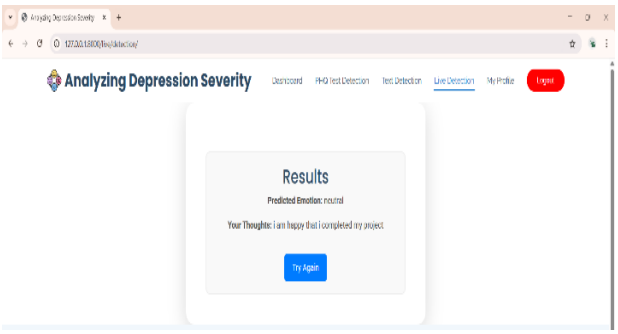

Fig: Facial Emotion & User Thoughts Prediction From Text

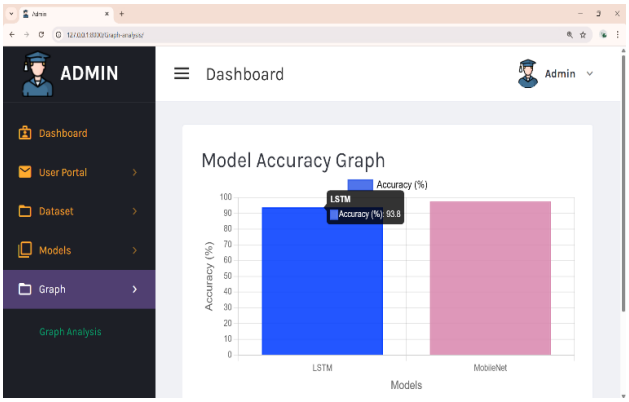

Fig: Model Accuracy Graph

VI. Conclusion & Future Scope

This study presents a multimodal AI-based system for depression detection, integrating PHQ-9, facial expression analysis, and NLP-based sentiment classification. The experimental results confirm the system's effectiveness in enhancing diagnostic accuracy. The proposed model achieves strong accuracy while remaining lightweight enough for real-time deployment.

Future enhancements include integrating:

- **Voice analysis** for tone and speech rate detection
- **Longitudinal tracking** for early relapse prevention
- **Mobile integration** for remote therapy tools
- **Explainable AI** techniques to increase trust and transparency

This paper presents a novel multimodal system for depression severity classification by fusing insights from facial emotion recognition, text sentiment analysis, and PHQ-9 responses. This research underscores the potential of deep learning in revolutionizing mental health diagnostics and making early intervention more accessible.

VII. REFERENCES

[1] World Health Organization. "Depression." 2023.

[2] M. Pantic et al., "Facial Expression Recognition for Mental Health Assessment," *Journal of Affective Computing*, vol. 12, no. 3, 2020.

[3] M. De Choudhury et al., "Social Media as a Predictor for Depression: A Machine Learning Approach," *ICWSM*, 2021.

[4] Devlin et al., "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *arXiv preprint arXiv:1810.04805*, 2019.

[5] A. Singh et al., "Multimodal Depression Detection Using Video and Text Analysis," *ACM Multimedia*, 2020.

[6] Kroenke K., Spitzer R. L., Williams J. B. W., "The PHQ-9: A Validated Tool for Depression Screening," *J Clin Psychiatry*, 2001.

[7] J. Liu et al., "Deep Learning for Depression Detection: A Survey," *Artificial Intelligence in Medicine*, vol. 129, 2022..